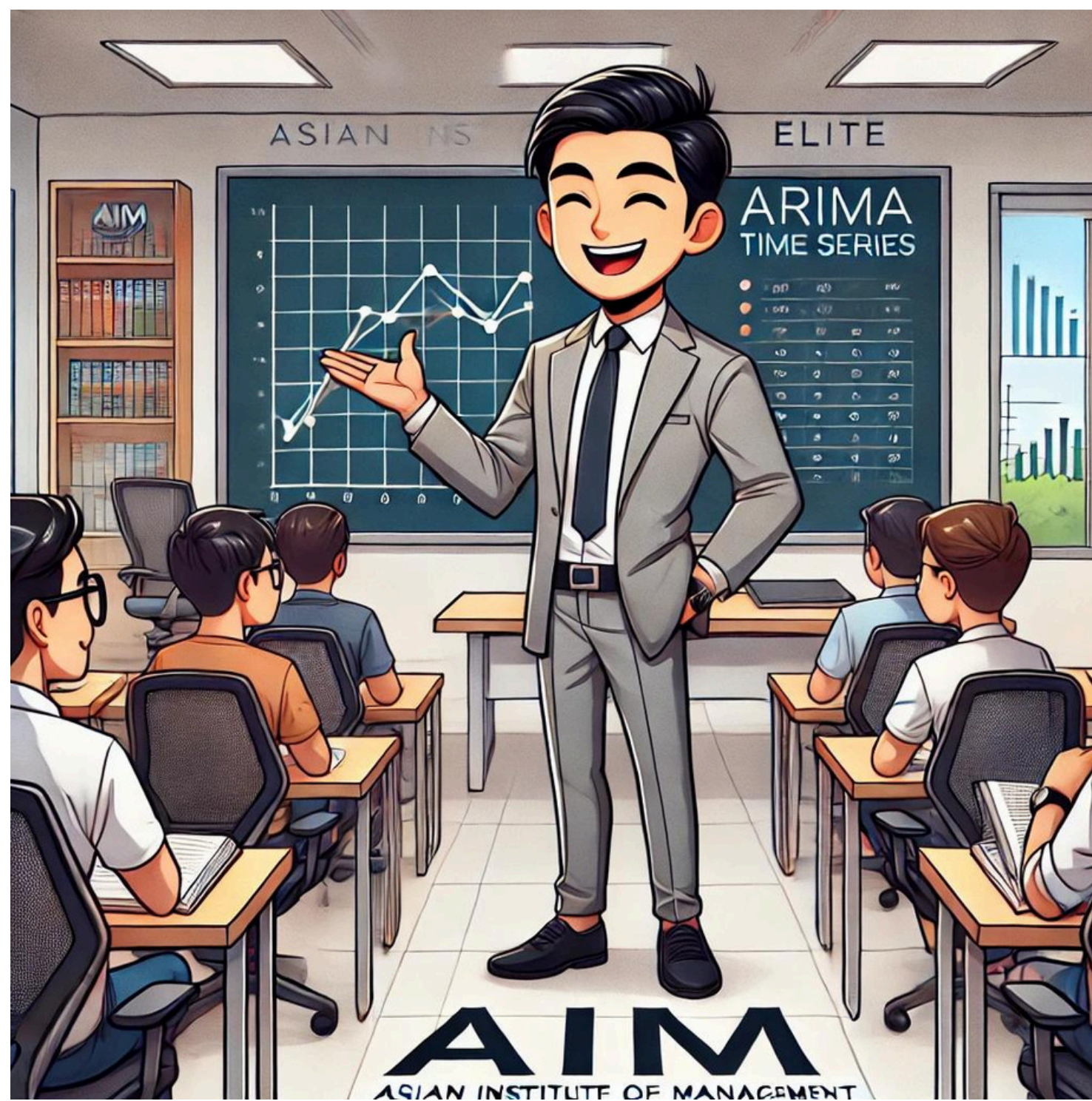




supplementary to Notebook 1

ARIMA

AutoRegressive Integrated Moving Average

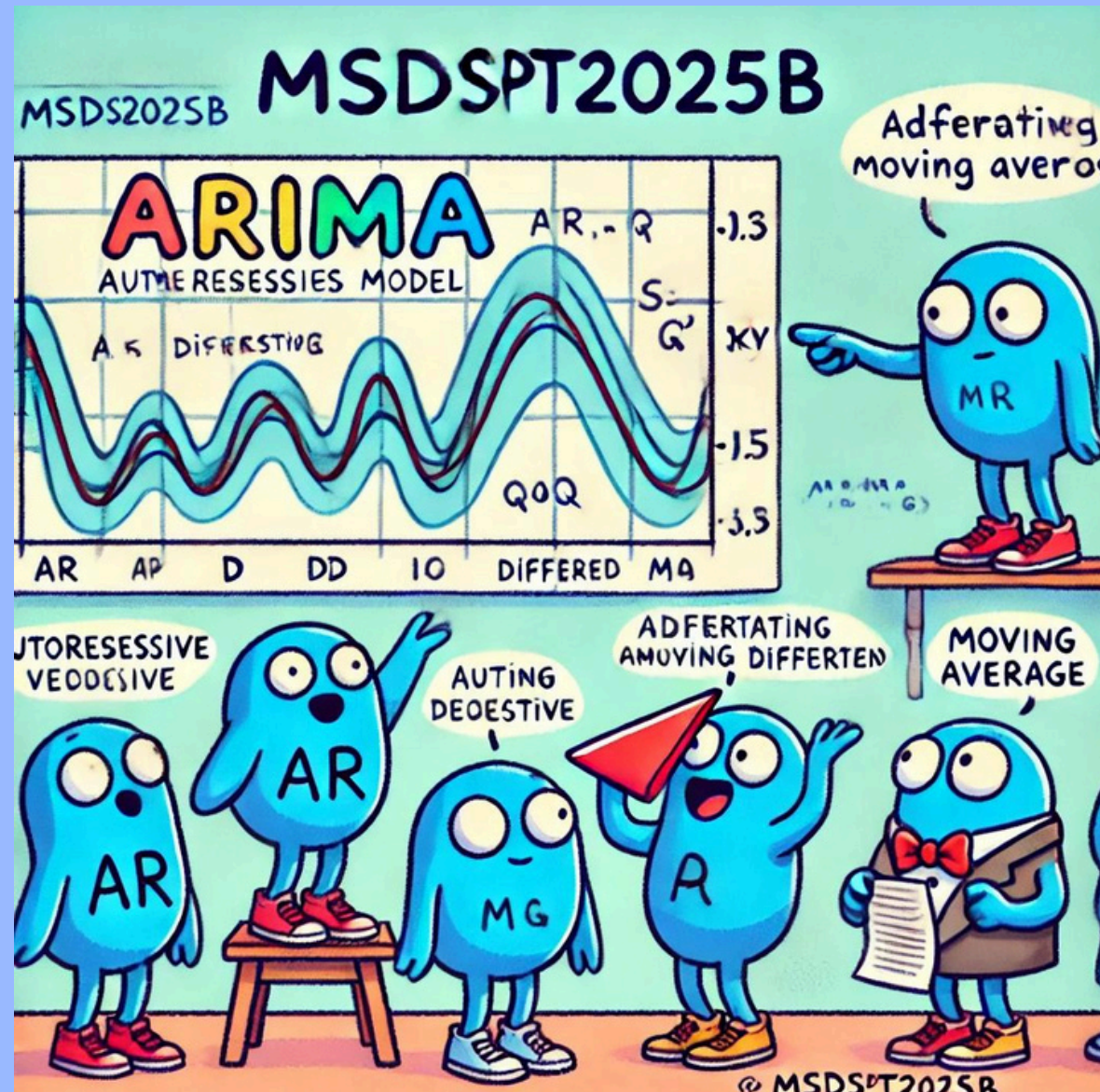
ML2 MSDSPT 2025B

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Aboitiz Chair in DS, Academician NAST

CONTENTS

Notebook 1



- 1 Algorithmic Foundation
- 2 Main Assumptions
- 3 Results: Champagne sales data

1. ALGORITHMIC FOUNDATION

AR: Autoregression. A model that uses the dependent relationship between an observation and some number of lagged observations.

ARIMA (1, 0, 0)

$$X_t = \mu + \alpha_1 X_{t-1}$$

ARIMA (2, 0, 0)

$$X_t = \mu + \alpha_1 X_{t-1} + \alpha_2 X_{t-2}$$

ARIMA (p, 0, 0)

$$X_t = \mu + \alpha_1 X_{t-1} + \alpha_2 X_{t-2} \cdots + \alpha_p X_{t-p}$$

ARIMA(1,0,0)) is a first-order autoregressive model: if the series is stationary and autocorrelated, perhaps it can be predicted as a multiple of its own previous value, plus a constant

1. ALGORITHMIC FOUNDATION

I: Integrated. The use of differencing of raw observations (e.g. subtracting an observation from an observation at the previous time step) in order to make the time series stationary.

$$\text{ARIMA (0, 0, 0)} \quad X_t = X_t$$

$$\text{ARIMA (0, 1, 0)} \quad X_t = X_{t-1} + \epsilon_t$$

$$\text{ARIMA (0, 2, 0)} \quad X_t = 2X_{t-1} + X_{t-2} + \epsilon_t + \epsilon_{t-1}$$

ARIMA(1,0,0)) is a first-order autoregressive model: if the series is stationary and autocorrelated, perhaps it can be predicted as a multiple of its own previous value, plus a constant

1. ALGORITHMIC FOUNDATION

MA: Moving Average. A model that uses the dependency between an observation and a residual error from a moving average model applied to lagged observations.

ARIMA (0, 0, q)
$$X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

where θ_q is the coefficient at time $t - q$ of the residual ε_{t-q}

ARIMA(1,0,0)) is a first-order autoregressive model: if the series is stationary and autocorrelated, perhaps it can be predicted as a multiple of its own previous value, plus a constant

1. ALGORITHMIC FOUNDATION

All together now! ARIMA! ARIMA!
AutoRegressive **I**ntegrated **M**oving **A**verage

Equations previously can be simplified as

$$X_t = \alpha_1 X_{t-1} + \cdots + \alpha_p X_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

$$X_t - \alpha_1 X_{t-1} - \cdots - \alpha_p X_{t-p} = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

Can be generalized and put in matrix form as:

ARIMA (p, d, q)

$$\left(1 - \sum_{i=1}^p \phi_i L^i\right) (1 - L)^d X_t = \delta + \left(1 + \sum_{i=1}^q \theta_i L^i\right) \varepsilon_t$$

Equation ready for maximum likelihood optimization : Newton Raphson or
Gradient Descent or Broyden-Fletcher-Goldfarb-Shanno (BFGS)

1. ALGORITHMIC FOUNDATION

All together now! **ARIMA** (p, d, q)
Auto**R**egressive **I**ntegrated **M**oving **A**verage

$$\left(1 - \sum_{i=1}^p \phi_i L^i\right) (1 - L)^d X_t = \delta + \left(1 + \sum_{i=1}^q \theta_i L^i\right) \varepsilon_t$$

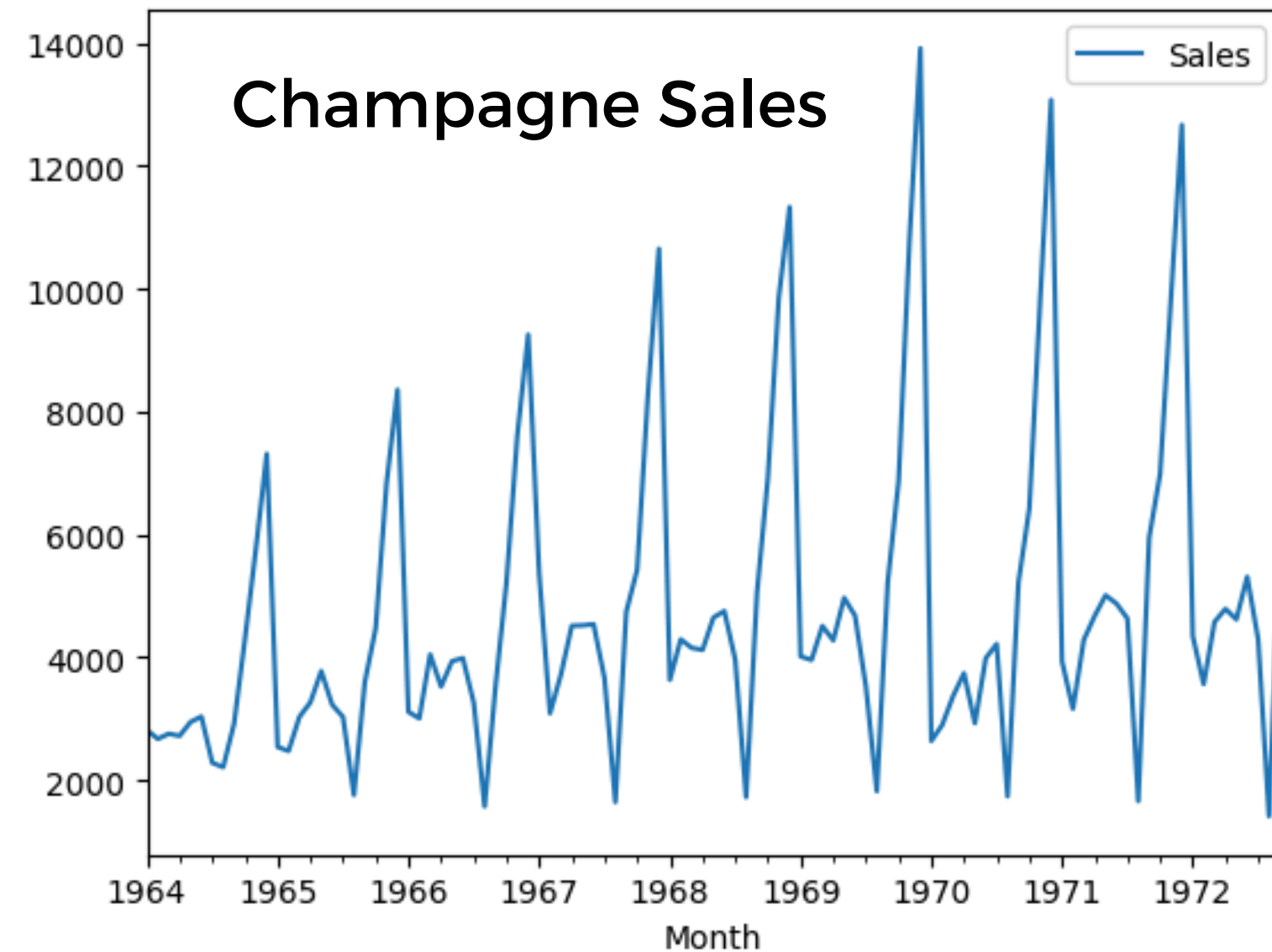
p: The number of lag observations included in the model, also called the lag order (deals with window of X_t)

d: The number of times that the raw observations are differenced, also called the degree of differencing (deals with order of differencing of X_t)

q: The size of the moving average window, also called the order of moving average (deals with residuals)

2. ASSUMPTIONS

- **ARIMA is the linear model** version of time-series forecasting, simplest ML baseline
- **Stationarity is critical** and we need to run statistics
- **Data to play with:**
Champagne Sales from 1964 to 1972



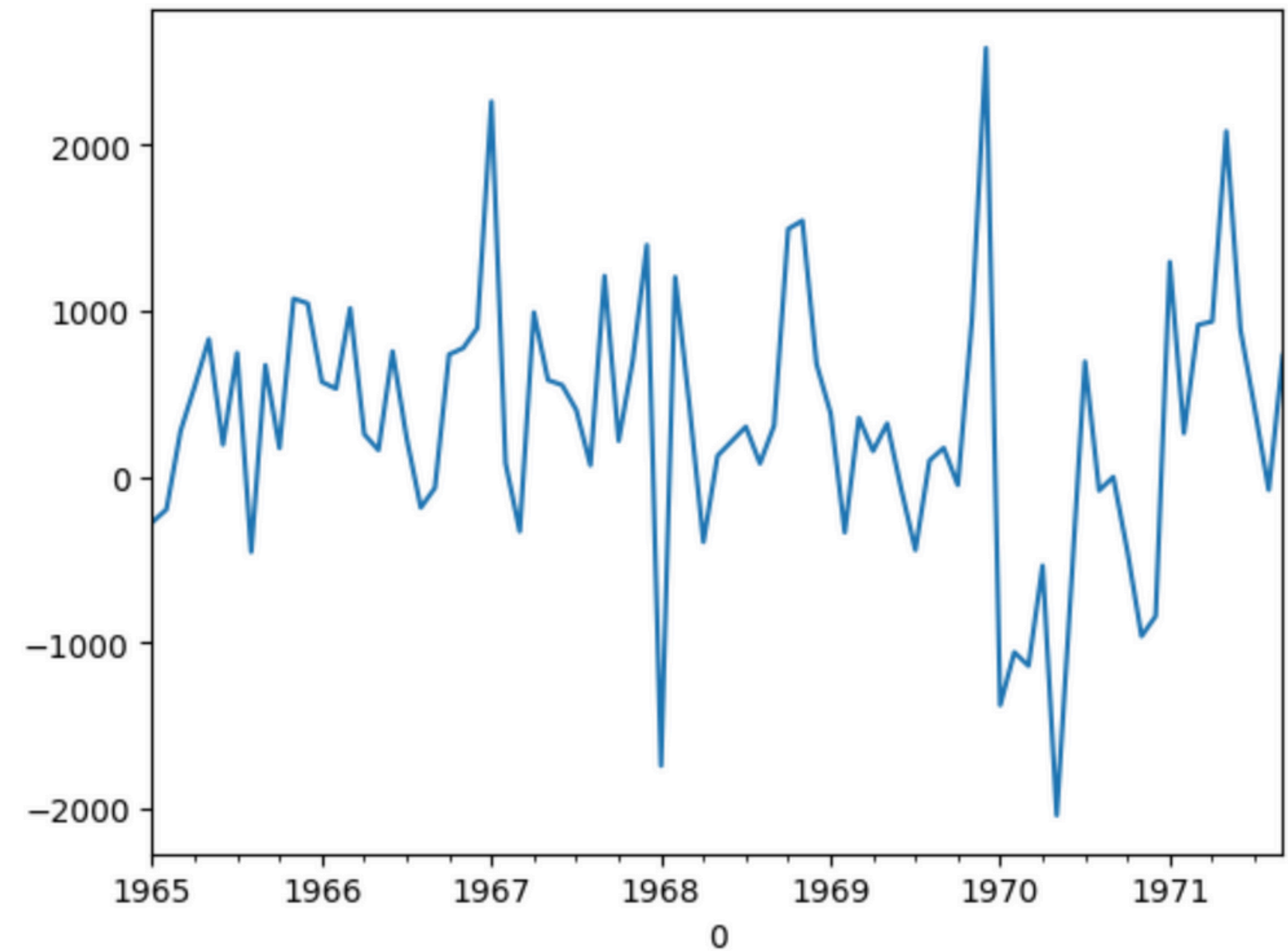
⇒ ADF Statistic: -1.445970
p-value: 0.560050
Critical Values:
1%: -3.514
5%: -2.898
10%: -2.586

Not Stationary! Needs differencing to pre-process for ARIMA.

2. ASSUMPTIONS

- **ARIMA is the linear model** version of time-series forecasting, simplest ML baseline
- **Stationarity is critical** and we need to run statistics
- **Data to play with:**
Champagne Sales from 1964 to 1972

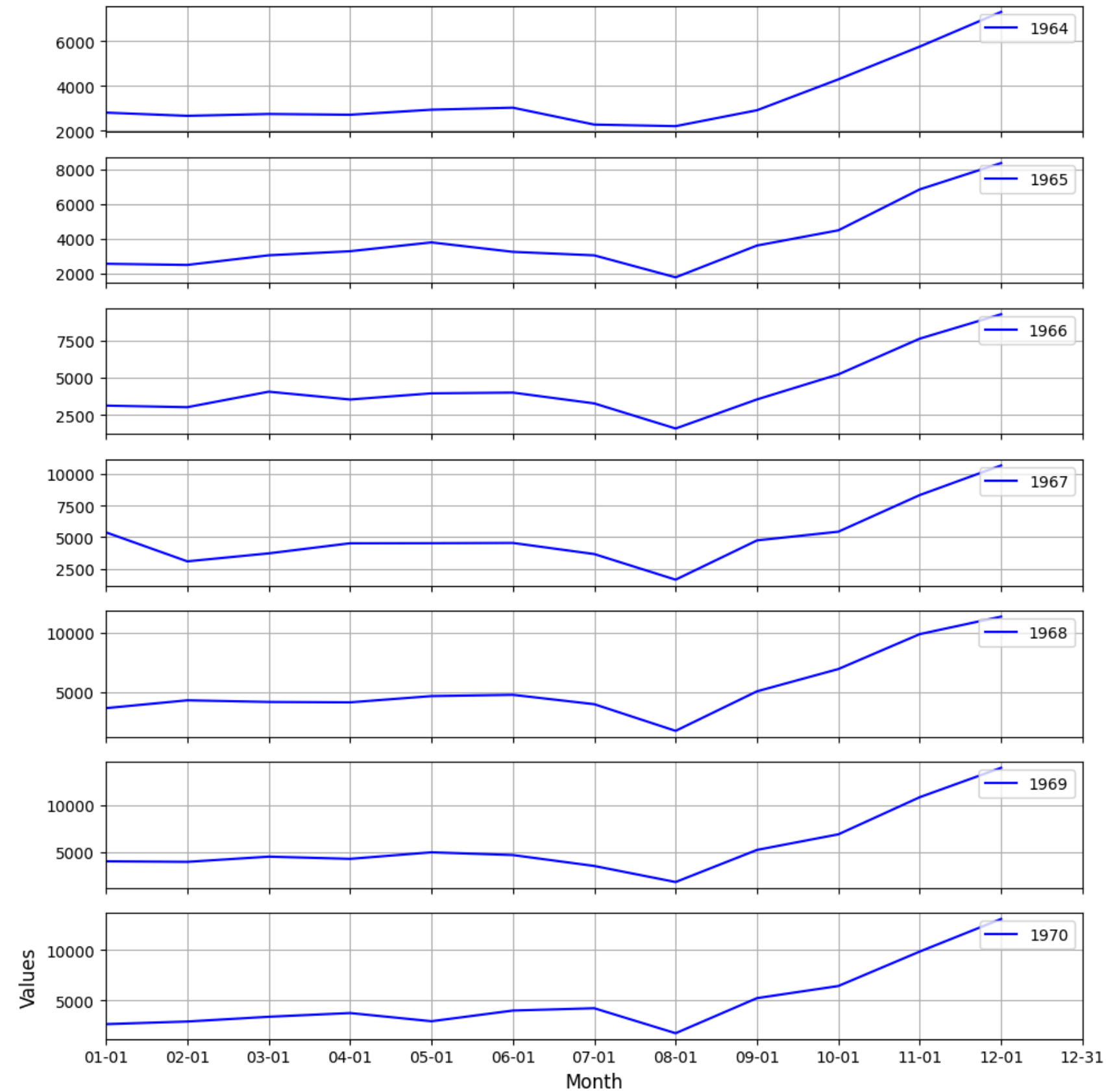
Champagne Sales difference (delta x)



⇒ ADF Statistic: -7.134898
p-value: 0.000000
Critical Values:
1%: -3.515
5%: -2.898
10%: -2.586

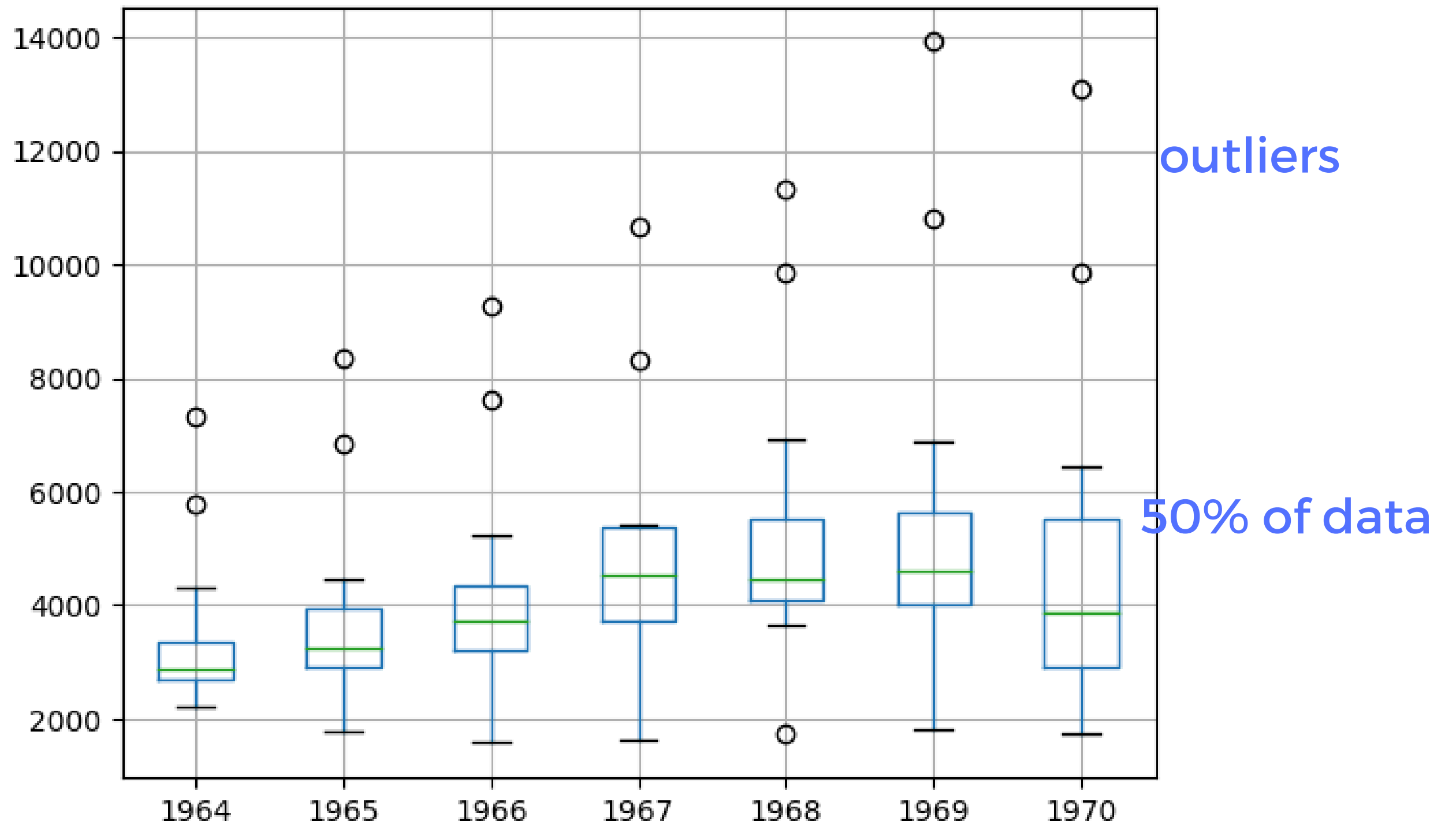
Stationary now after one time differencing

3. RESULTS: EDA



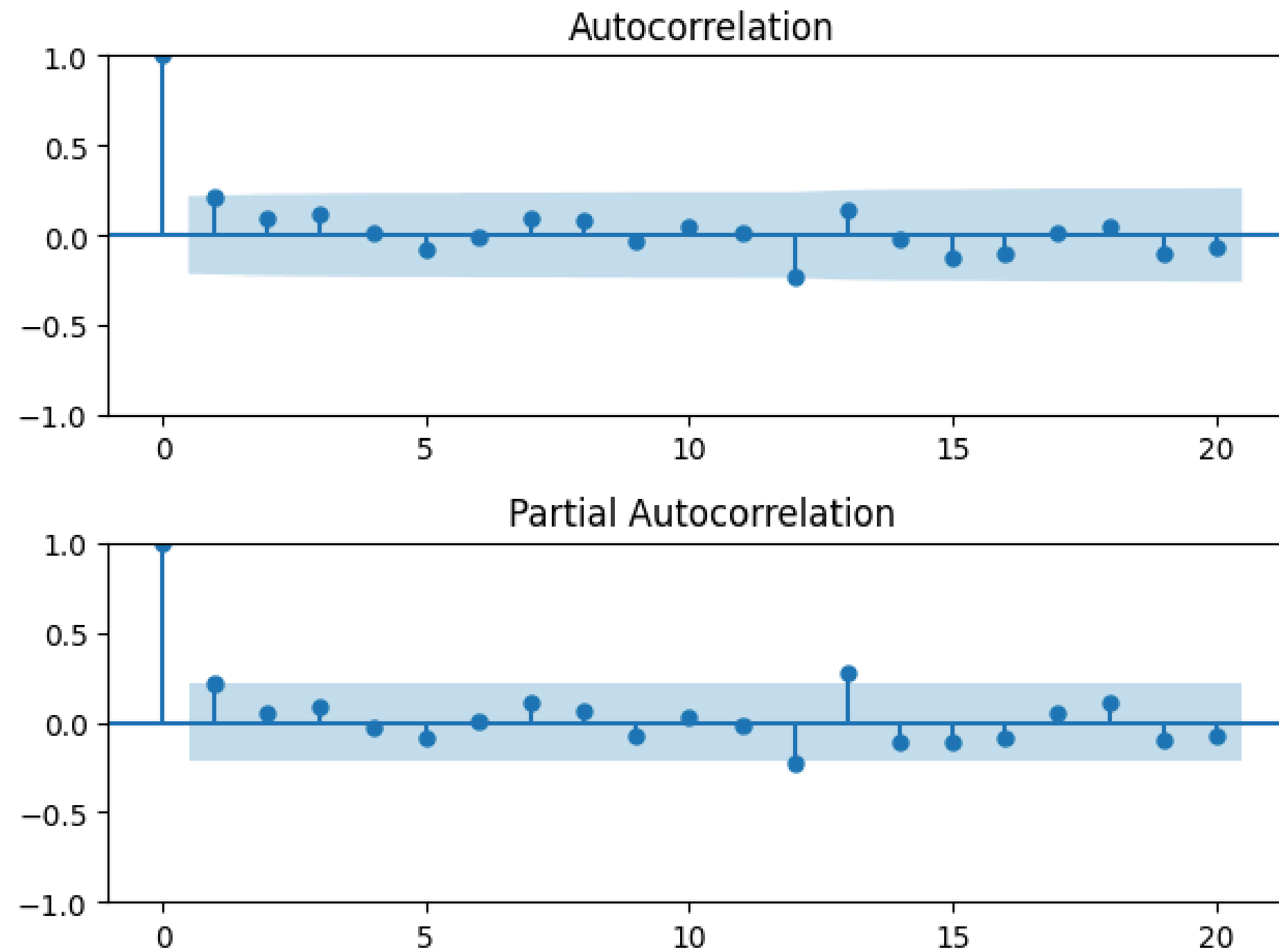
year-on-year alignment: We observed seasonality

3. RESULTS: EDA



Box and whiskers plot: Overall, Increasing trend

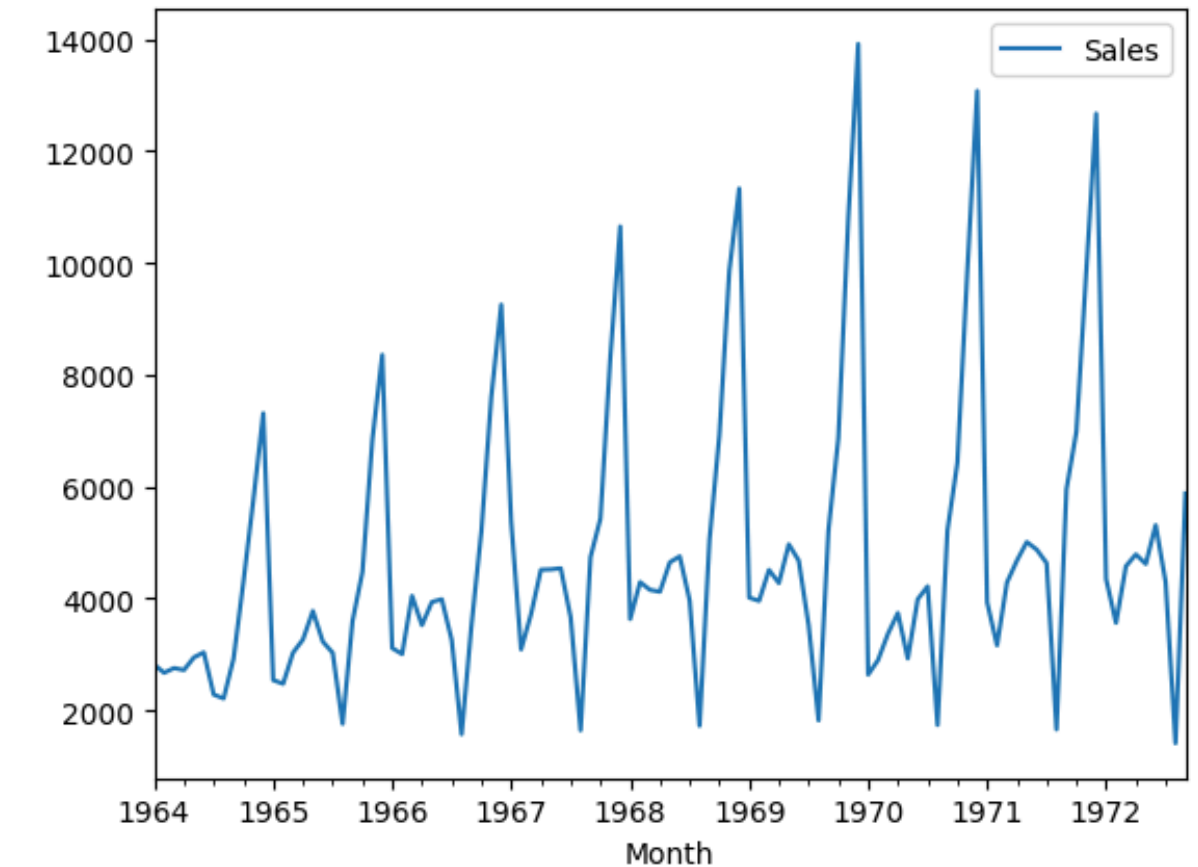
3. RESULTS: EDA



Significant lag for 1 month and 12, 13th month

3. RESULTS: BASELINE

- If we assume the next time step is just equal to the current time step:
RMSE = 3167 MAE=2100
- **2100** is better than std of **2486**, so not so bad a baseline!



Statistics of Champagne Sales

```
➡ count      93.000000
   mean    4641.118280
   std    2486.403841
   min    1573.000000
   25%    3036.000000
   50%    4016.000000
   75%    5048.000000
   max   13916.000000
   Name: 1, dtype: float64
```

3. RESULTS: ARIMA RESULTS

```
# Define ranges for p, d, q
p_range = range(0, 4)
d_range = range(0, 2)
q_range = range(0, 4)

# Initialize a dictionary to store the results
results = {}

# Iterate over all combinations of p, d, q
for p, d, q in product(p_range, d_range, q_range):
    order = (p, d, q)
    # Evaluate the ARIMA model for the current (p, d, q) order
    rmse = evaluate_arima_model(X, order)
    # Store the result in the dictionary
    results[order] = rmse
    print(f"Order {order}, RMSE: {rmse}")

# Find the (p, d, q) combination with the lowest RMSE
best_order = min(results, key=results.get)
best_rmse = results[best_order]

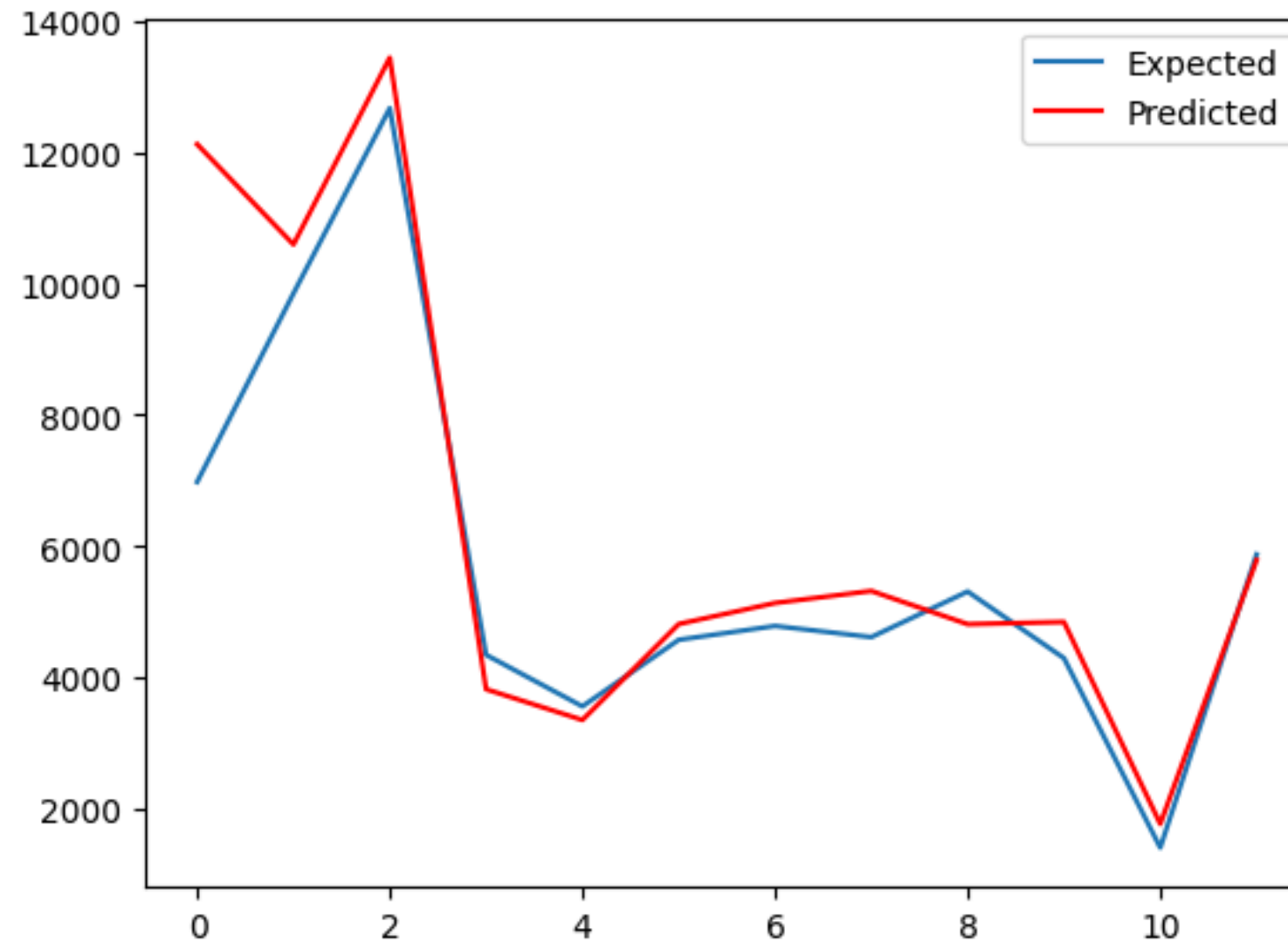
# Print the best (p, d, q) combination and the corresponding RMSE
print(f"\nBest Order: {best_order}, Lowest RMSE: {best_rmse}")
```

```
Order (0, 0, 0), RMSE: 1013.4644053594292
Order (0, 0, 1), RMSE: 971.7597612339258
Order (0, 0, 2), RMSE: 1043.4536201042029
Order (0, 0, 3), RMSE: 1060.4957312872298
Order (0, 1, 0), RMSE: 956.4557381015435
Order (0, 1, 1), RMSE: 1031.0959046110572
Order (0, 1, 2), RMSE: 988.3614342018857
Order (0, 1, 3), RMSE: 1049.8509711389418
Order (1, 0, 0), RMSE: 967.1040144683225
Order (1, 0, 1), RMSE: 1038.5151529676098
Order (1, 0, 2), RMSE: 1074.584785880845
Order (1, 0, 3), RMSE: 1084.2876059658352
Order (1, 1, 0), RMSE: 910.7211044976348
Order (1, 1, 1), RMSE: 996.9211294668796
Order (1, 1, 2), RMSE: 1034.1172420966816
Order (1, 1, 3), RMSE: 1037.5349038954344
Order (2, 0, 0), RMSE: 1029.1640725012937
Order (2, 0, 1), RMSE: 1093.3067379821418
Order (2, 0, 2), RMSE: 1090.2151524790784
Order (2, 0, 3), RMSE: 1068.3256130300185
Order (2, 1, 0), RMSE: 1009.7903676244889
Order (2, 1, 1), RMSE: 1036.5359136739191
Order (2, 1, 2), RMSE: 1052.72894057704
Order (2, 1, 3), RMSE: 1130.1748632758558
Order (3, 0, 0), RMSE: 1050.2875267026443
Order (3, 0, 1), RMSE: 1087.2915988728737
Order (3, 0, 2), RMSE: 1091.3689152696152
Order (3, 0, 3), RMSE: 1061.4119909946376
Order (3, 1, 0), RMSE: 1000.6789356950111
Order (3, 1, 1), RMSE: 1045.224939023802
Order (3, 1, 2), RMSE: 1051.6704317928163
Order (3, 1, 3), RMSE: 1033.384707062128
```

Best Order: (1, 1, 0), Lowest RMSE: 910.7211044976348

the best configuration discovered was **ARIMA(1, 1, 0)** with an **RMSE of 910** way better than baseline **RMSE ~3167**

3. RESULTS: ARIMA RESULTS, ROLLING FORECAST FOR 12 MONTHS



Not Bad! But this is a simple data and most of you are not in champagne forecasting business :)

[Let's now browse Notebook 1](#)

SUMMARY: KEY ASSUMPTIONS FOR ARIMA

- **Stationarity:** ARIMA assumes that the time series is stationary, meaning that its statistical properties (mean and variance) do not change over time. If the series is not stationary, differencing (determined by the d parameter) is applied to make it stationary, but making it stationary in the first place is often better.
- **Linearity:** ARIMA assumes a linear relationship between past values and future values. The model relies on linear combinations of past observations (AR terms) and past forecast errors (MA terms) to predict future values.
- **No significant seasonal effects:** Standard ARIMA does not account for seasonality. If the time series has a **seasonal pattern**, you need to **extend ARIMA to Seasonal ARIMA (SARIMA)** by incorporating seasonal terms to model the periodic fluctuations.

[Let's now browse Notebook 1](#)